

Analysis Report 3

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```
football <- read.csv("C:/Users/mandu/Downloads/2020-2021.csv")
```

Data Description

The data I am using for this report is titled “English Premier League” and comes from kaggle.com. The dataset contains 20 years worth of information and statistics about football matches played in the English Premier League. The dataset contains a variety of data including the home and away teams, goals, red cards, audience attendance, and more. There is also a lot of data for sports betting purposes, such as gambling odds data from the gambling website bet365.

Research Question

The guiding research question for the report is: “Does playing at home significantly increase the odds of winning a football match?” This is a useful question to investigate with regards to predicting the outcome of future football matches. A real-world use for the results of the investigation could be for better odds in sports betting.

Variables of Interest

There are many variables that can be investigated, but for the purposes of this report, the following are the variables I am interested in:

HTHG - This variable is an integer that represents the number of half time home team goals.

HTAG - This variable is an integer that represents the number of half time away team goals.

B365H - This variable is a float (decimal number) that represents the home team win odds based on what the gambling site bet365 says.

B365A - This variable is a float that represents the away team win odds based on what the gambling site bet365 says.

In order to investigate the research question, I will also be creating a new variable:

HWIN - This is an integer variable I am creating which will represent whether the home team won or not (stored as 1 or 0)

In order to build this variable, we will be using this variable from the dataset:

FTR - This is a character variable which stands for Full Time Result and reflects whether there was home win, away win, or draw

Data Wrangling

In this section I will make my new variable HWIN and remove missing values.

#First, the new variable

```
football$HWIN <- ifelse(football$FTR == "H", 1, 0)
```

#Removing missing values

```
football_clean <- football[complete.cases(football[, c("HTHG", "HTAG",  
                                                    "B365H", "B365A"))], ]
```

Now we have a football_clean dataset which should only contain present values, and we have our new HWIN variable which represents home wins with a 1 and home losses or draws with a 0.

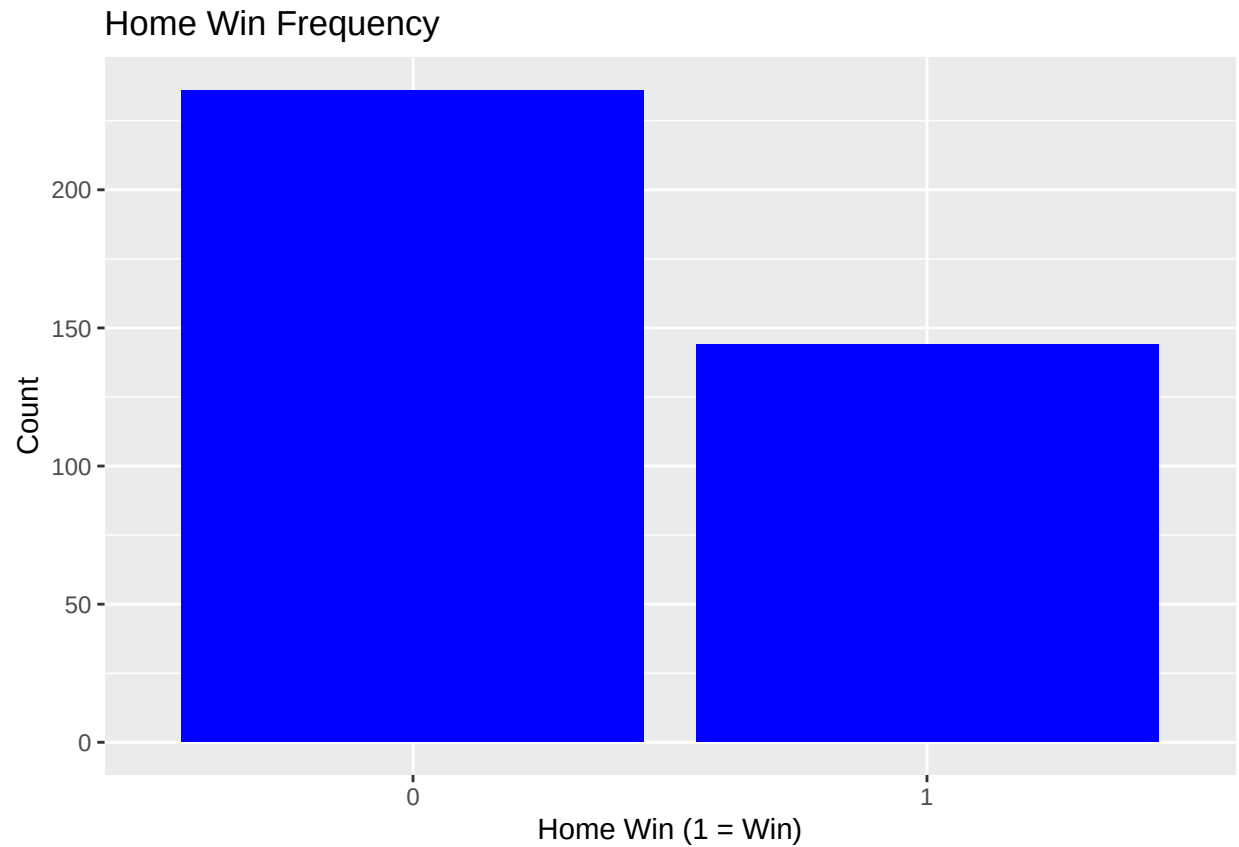
EDA

We will investigate win frequency (HWIN) individually and then look at how each variable compares to HWIN.

HWIN:

```
library(ggplot2)

ggplot(football_clean, aes(x = factor(HWIN))) +
  geom_bar(fill = "blue") +
  labs(title = "Home Win Frequency",
       x = "Home Win (1 = Win)",
       y = "Count")
```

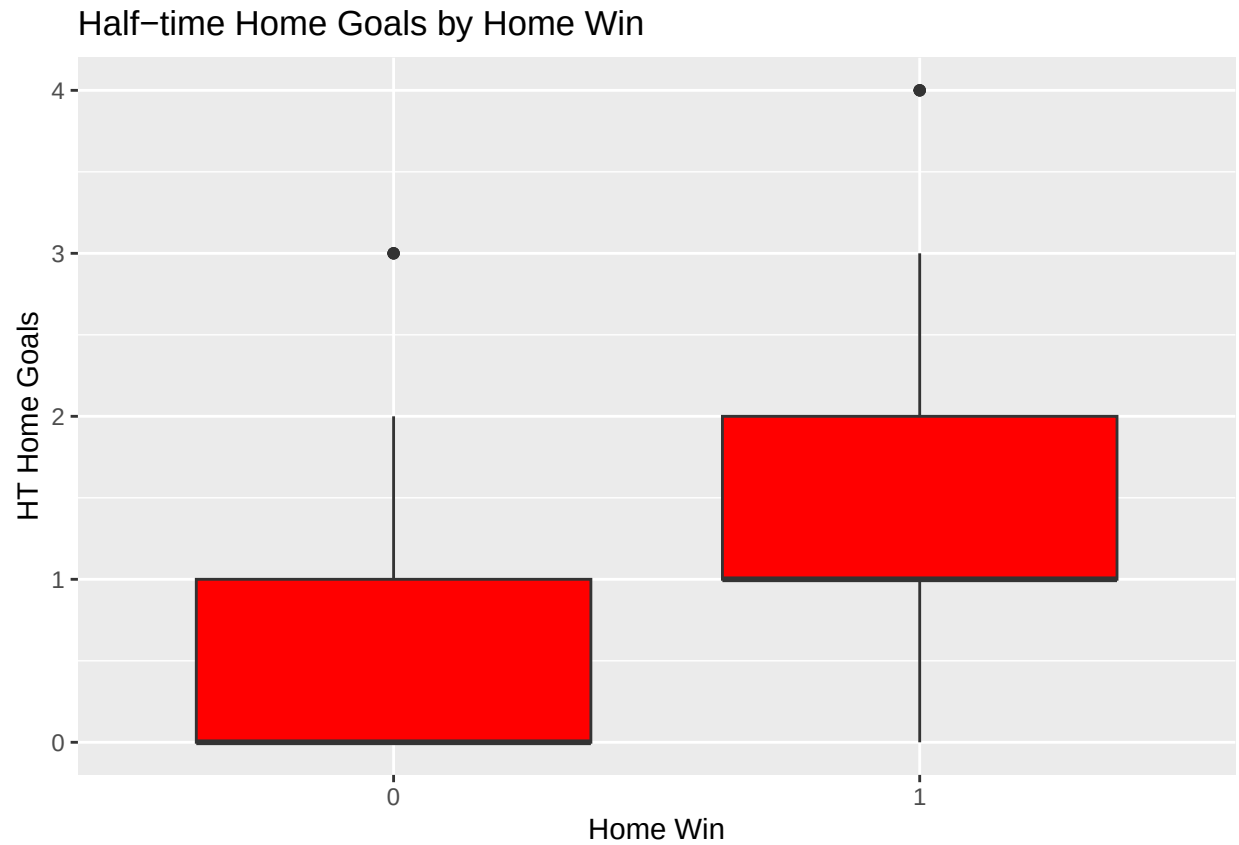


Here we can see that the home wins are actually lower than the away wins and draws combined.

Now lets look at the variables compared to HWIN.

Half-time Home Goals vs Home Win:

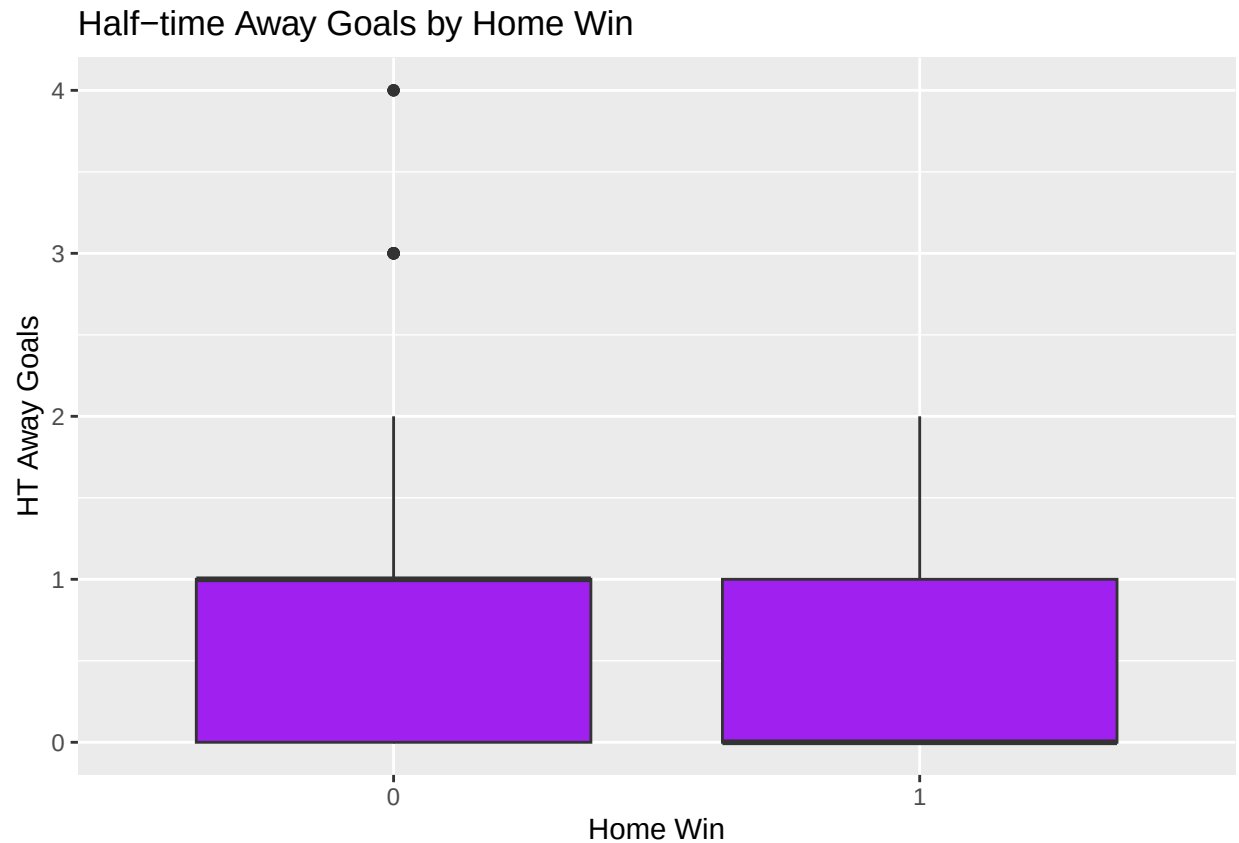
```
ggplot(football_clean, aes(x = factor(HWIN), y = HTHG)) +  
  geom_boxplot(fill = "red") +  
  labs(title = "Half-time Home Goals by Home Win",  
        x = "Home Win",  
        y = "HT Home Goals")
```



As one would expect, half-time home goals were greater during home win games.

Half-time Away Goals vs Home Win:

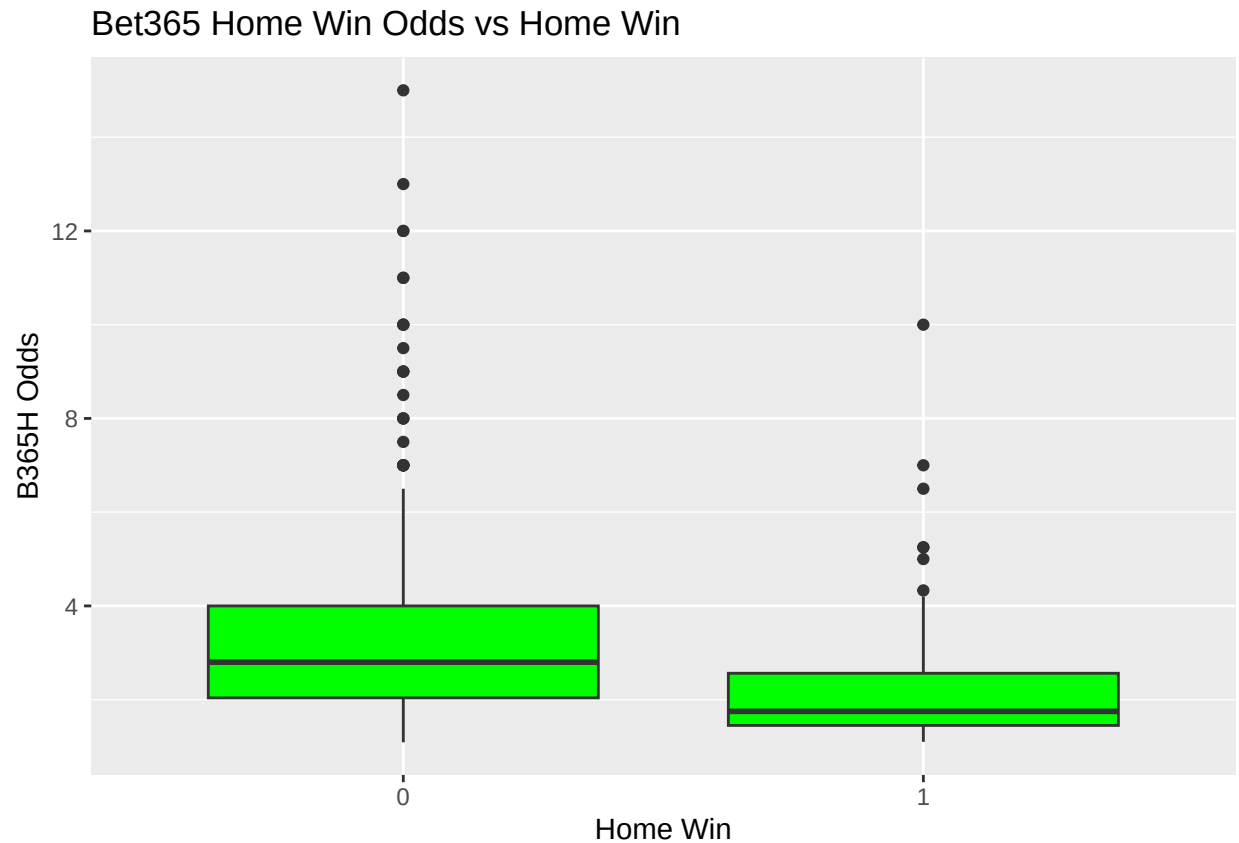
```
ggplot(football_clean, aes(x = factor(HWIN), y = HTAG)) +  
  geom_boxplot(fill = "purple") +  
  labs(title = "Half-time Away Goals by Home Win",  
        x = "Home Win",  
        y = "HT Away Goals")
```



Interestingly half-time away goals are typically similar in both home win and non home win games, except there are notable outliers during home loss/draw games.

Betting Odds for Home Win vs Home Win

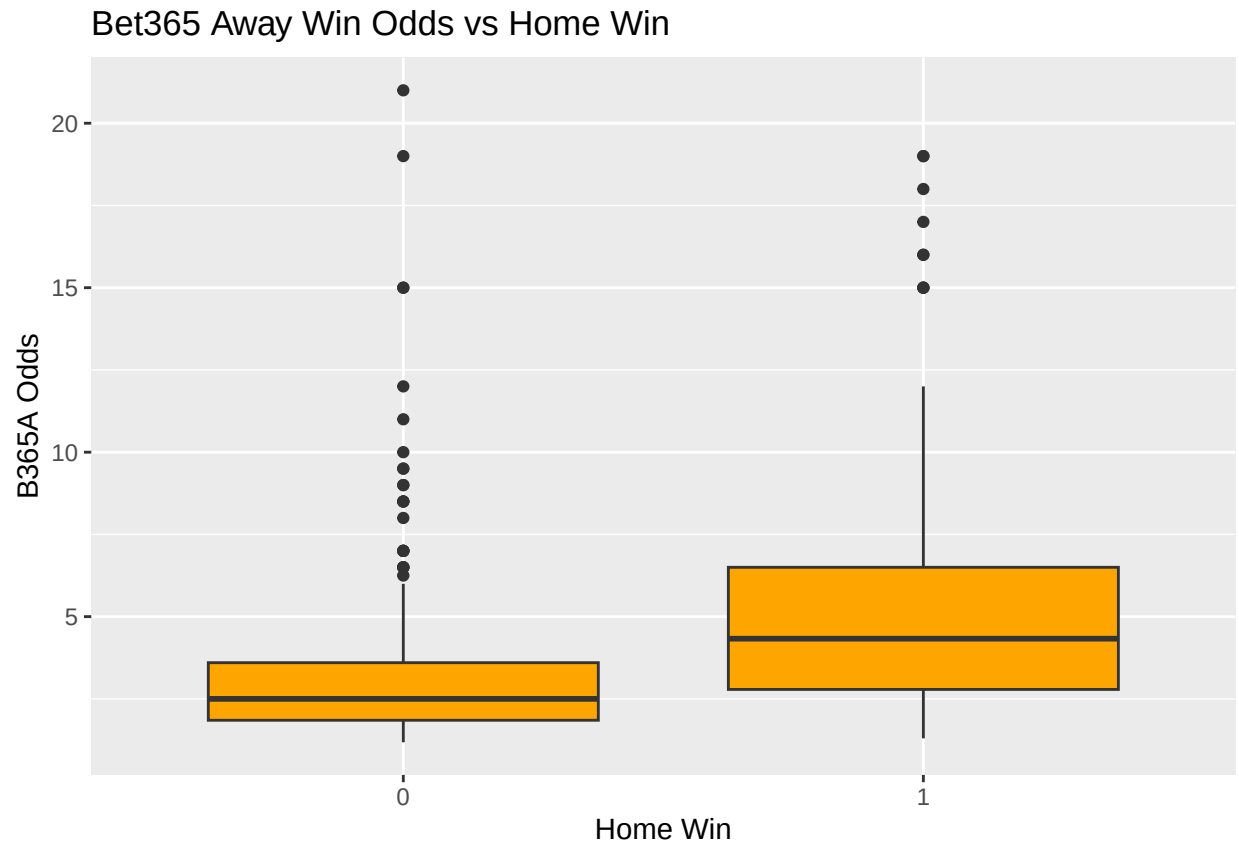
```
ggplot(football_clean, aes(x = factor(HWIN), y = B365H)) +  
  geom_boxplot(fill = "green") +  
  labs(title = "Bet365 Home Win Odds vs Home Win",  
        x = "Home Win",  
        y = "B365H Odds")
```



It looks like when the home team wins, the bet365 odds tend to be lower spread suggesting that they were expected to win (supports the predictive validity of betting odds)

Betting Odds for Away Win vs Home win

```
ggplot(football_clean, aes(x = factor(HWIN), y = B365A)) +  
  geom_boxplot(fill = "orange") +  
  labs(title = "Bet365 Away Win Odds vs Home Win",  
        x = "Home Win",  
        y = "B365A Odds")
```



Similarly, when the away team wins the away win odds are also less spread which is consistent with the established predictive power of the bet365 odds data.

Model Selection

For this report I will be building a logistic regression model. The HWIN variable is binary, and the logistic regression model is ideal for binary classification problems. Since I am not modeling for draw outcomes, only wins or non-wins, there is no need for a multinomial regression.

Model Building

For the initial model, I will first see how halftime goals specifically influence match outcome.

Model 1:

```
model1 <- glm(HWIN ~ HTHG + HTAG,
              data = football_clean, family = binomial)
```

```
summary(model1)
```

```
##
## Call:
## glm(formula = HWIN ~ HTHG + HTAG, family = binomial, data = football_clean)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
```

```
## (Intercept) -0.9243      0.1860  -4.969 6.74e-07 ***
## HTHG         1.5986      0.2024   7.896 2.87e-15 ***
## HTAG        -1.3558      0.2279  -5.950 2.68e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 504.30  on 379  degrees of freedom
## Residual deviance: 366.92  on 377  degrees of freedom
## AIC: 372.92
##
## Number of Fisher Scoring iterations: 5
```

Both HTHG and HTAG have p values that are very small so they are both statistically significant.

Since we have established the predictive power of the bet365 odds in the EDA section, we can add all the variables in the second model and will likely see all variables be significant.

Model 2:

```
model2 <- glm(HWIN ~ HTHG + HTAG + B365H + B365A,
              data = football_clean, family = binomial)

summary(model2)

##
## Call:
## glm(formula = HWIN ~ HTHG + HTAG + B365H + B365A, family = binomial,
##      data = football_clean)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.07649    0.51019   0.150 0.880830
## HTHG         1.55941    0.20615   7.565 3.89e-14 ***
## HTAG        -1.27024    0.23826  -5.331 9.75e-08 ***
## B365H        -0.44020    0.12876  -3.419 0.000629 ***
## B365A         0.03688    0.05275   0.699 0.484412
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 504.30  on 379  degrees of freedom
## Residual deviance: 336.06  on 375  degrees of freedom
## AIC: 346.06
##
## Number of Fisher Scoring iterations: 6
```

After adding the betting odds, it seems that B365A is the only not statistically significant variable with a p value of 0.484, while the other variables' p values are much smaller.

It is possible that B365A is a redundant variable and that B365H captures all relevant info from the variable anyways. Since it is not a significant variable, let's remove it for our third model then run diagnostics.

Model 3:


```

model3 <- glm(HWIN ~ HTHG + HTAG + B365H,
              data = football_clean, family = binomial)

summary(model3)

##
## Call:
## glm(formula = HWIN ~ HTHG + HTAG + B365H, family = binomial,
##      data = football_clean)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   0.3663     0.3059   1.197   0.231
## HTHG          1.5865     0.2038   7.786 6.90e-15 ***
## HTAG         -1.2733     0.2376  -5.359 8.38e-08 ***
## B365H        -0.4962     0.1045  -4.750 2.04e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 504.30  on 379  degrees of freedom
## Residual deviance: 336.56  on 376  degrees of freedom
## AIC: 344.56
##
## Number of Fisher Scoring iterations: 5

```

Now, lets run some diagnostics to see if any more changes need to be made to the model.

Checking for multicollinearity:

```

library(car)

## Loading required package: carData

vif(model3)

##      HTHG      HTAG      B365H
## 1.104016 1.066485 1.037164

```

Since all the VIF values are under 5, there does not seem to be any multicollinearity among the variables.

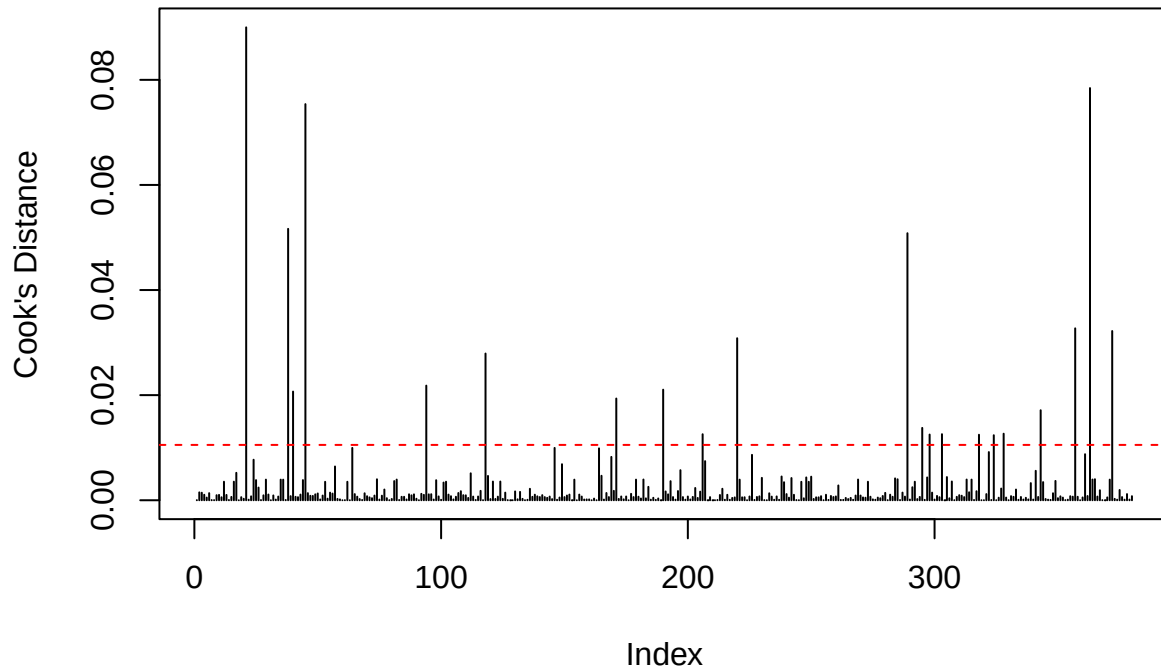
Checking for outliers:

```

plot(cooks.distance(model3), type = "h",
     main = "Cook's Distance", ylab = "Cook's Distance")
abline(h = 4/nrow(football_clean), col = "red", lty = 2)

```

Cook's Distance



Based on this Cook's distance plot, there are a few outliers above the red threshold line, but none of them appear extreme with the highest only reaching around 0.09.

Since it seems that multicollinearity is not present, and the outliers are not significantly affecting our results, I am satisfied with presenting model3 as our final model:

```
summary(model3)
```

```
##
## Call:
## glm(formula = HWIN ~ HTHG + HTAG + B365H, family = binomial,
##      data = football_clean)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   0.3663     0.3059   1.197   0.231
## HTHG           1.5865     0.2038   7.786 6.90e-15 ***
## HTAG          -1.2733     0.2376  -5.359 8.38e-08 ***
## B365H         -0.4962     0.1045  -4.750 2.04e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 504.30  on 379  degrees of freedom
## Residual deviance: 336.56  on 376  degrees of freedom
## AIC: 344.56
```

```
##  
## Number of Fisher Scoring iterations: 5
```

Analytic Interpretations

The final logistic regression model includes half-time home goals, half-time away goals, and bet365 odds for home win in order to predict an outcome variable of home game wins. All three variables were found to be statistically significant at the 0.05 level with these values:

HTHG - std error: 0.203, p value: < 0.001 HTAG - std error: 0.237, p value: < 0.001 B365H - std error: 0.104, p value: < 0.001

By converting the coefficients to odds ratios, we can interpret the variables:

```
#HTHG  
exp(1.58)
```

```
## [1] 4.854956
```

```
#HTAG  
exp(-1.27)
```

```
## [1] 0.2808316
```

```
#B365H  
exp(-0.49)
```

```
## [1] 0.6126264
```

Based on the odds ratios:

Holding other variables constant, for each additional goal the home team scores by half-time, the odds of a home win increase by a factor of ~4.85. For each additional goal the away team scores at half-time, the odds of a home win are reduced by ~72%. A 1 unit increase in the odds against the home team winning corresponds to a ~39% decrease in the odds of a home win.

To summarize:

More half time home goals greatly improve win chances, more half time away goals reduces home win odds, and lower bet365 odds for a home win (higher expectation for success) correspond to the actual home wins.

Classification Evaluation

To convert predicted probabilities into binary classifications (home win vs non home win), I used a common neutral threshold of 0.5:

```
pred_prob <- predict(model3, type = "response")  
pred_class <- ifelse(pred_prob >= 0.5, 1, 0)
```

With these classifications, I will build a confusion matrix:

```
table(Predicted = pred_class, Actual = football_clean$HWIN)
```

```
##           Actual  
## Predicted    0    1  
##           0 212  52  
##           1  24  92
```

Based on the confusion matrix, here are the performance metrics:

Accuracy = (TP + TN) / Total = (92 + 212) / (212 + 52 + 24 + 92) = 304 / 380 = 80%

$$\text{Sensitivity} = \text{TP} / (\text{TP} + \text{FN}) = 92 / (92 + 52) = 92 / 144 = 63.9\%$$

$$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP}) = 212 / (212 + 24) = 212 / 236 = 89.8\%$$

After comparing the model to the actual dataset:

The model is strong at identifying when the home team does not win with nearly a 90% specificity. However, the model is less sensitive to actual home wins with a recall of almost 64%, which means the model may be conservative and underpredict wins. With an overall accuracy of 80%, this simple logistic regression model is fairly accurate at making predictions.

Discussion

To answer the question of “Does playing at home significantly increase the odds of winning a football match?” I created a logistic regression model with the variables of half time home goals, half time away goals, and betting odds. The final model suggests that home goals significantly increase the odds of a home win while away goals reduce the odds of a home win, and higher odds (less favorable) are associated with lower chances to win. The model was especially accurate at determining when the home team would not win with an overall accuracy of 80%. Playing at home does appear to increase the odds of winning a football match according to my findings.

Some limitations to this research include that only half time data was used, which obviously only captures the first half of an entire football game. Additionally, betting odds may be biased as they are based on markets and could not be entirely indicative of team quality.

For future research, one could incorporate full match statistics, and perhaps explore outside factors affecting home games such as weather, season, or even the day of the week. It could also be informative to see how results change when comparing teams with known histories or rivalries.